

Competency

- To solve the real life problems based on linear simultaneous equations

Learning Outcomes

- To solve the linear simultaneous linear equations by substitution, elimination and graphical methods

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define linear equation - To define simultaneous linear equations - To tell the methods of solving pair of linear equations
2.	Understanding (U)	- To solve the pair of simultaneous equation (simple) by substitution method - To solve the pair of simple simultaneous equation by elimination method
3.	Application (A)	- To solve the pair of simultaneous equation by substitution method - To solve the pair of simultaneous equation by elimination method - To make the equation for the given conditions and solve them
4.	High Ability (HA)	- To identify the contextual problems as related to simultaneous equations, make the equations and solve it

Required Teaching Materials/ Resources

Graph board, graph paper, charts with various pair of linear equations, Geo-gebra tool, list of grocery items with rates of costs, audio-video materials etc.

Pre-knowledge: Linear equation, solving simultaneous equations by graphical method

Teaching Activities

1. Warm up the class with some mathematical logics, quiz questions, game etc.
2. Divide the students into four five groups and give some real life problems (if possible with pictures) to solve as fast as possible

Group A: The cost of a goat is five times the cost of a hen. If their total cost is Rs 12000, find the cost of each.

Group B: The mother is thrice as old as her son. If their total age is 40 years, find their present ages.

Group C: The cost of a pant is Rs 500 more than that of the shirt. If the total cost of the shirt and pant is Rs 1600, find the cost of each.

Group D: The cost of umbrella is Rs 300 less than that of a bag. If the cost of two umbrellas and a bag is Rs 1500, find the cost of each item
3. Discuss about their equations that the students made, solutions and methods of solving the problems in the above activities
4. Discuss about the substitution method with proper examples
5. Explain about the elimination method with proper examples
6. Discuss about the graphical method with proper examples
7. Give the values of variables x and y and tell the students to make the equations satisfying those values and discuss the solutions in the class

1. Solve the following simultaneous equations graphically.

a) $x + 2y = 14$ and $x - y = 5$

b) $\frac{x-2}{2} = y = \frac{2x-6}{3}$

Solution:

a) The given linear equation are

$x + 2y = 14$ (i)

and $x - y = 5$ (ii)

Now from equation (i).

$x + 2y = 14$

or, $y = \frac{14-x}{2}$

x	0	2	4
y	7	6	5

Plotting the point (0,7), (2,6) and (4,5) and joining them to form a straight line.

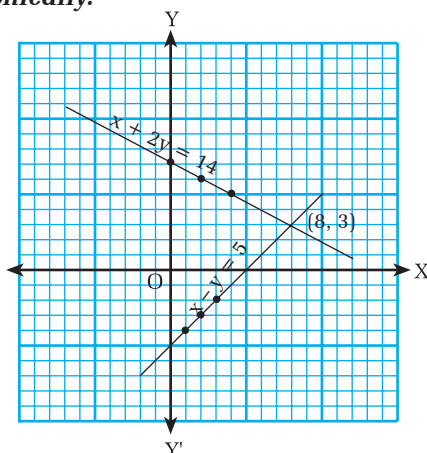
Also, from equation (ii), $x - y = 5 \therefore y = x - 5$

x	1	2	3
y	-4	-3	-2

Plotting the point (1,-4), (2,-3) and (3,-2) and joining them to form a straight line.

Since the graphs of equation $x + 2y = 14$ and $x - y = 5$ intersect at (8,3).

So, $x = 8$ and $y = 3$



b) The given linear equation are

$\frac{x-2}{2} = y$ (i)

and $y = \frac{2x-6}{3}$ (ii)

Now, from equation (i), $y = \frac{x-2}{2}$

x	0	2	4
y	-1	0	1

Plotting the point (0,-1), (2,0) and (4,1) and joining them to form a straight line.

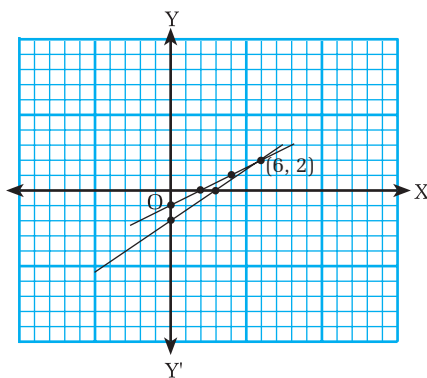
Also, from equation (ii), $y = \frac{2x-6}{3}$

x	0	3	6
y	-2	0	2

Plotting the point (1,-2), (3,0) and (6,2) and joining them to form a straight line.

Since the graphs of equation $y = \frac{x-2}{2}$ intersect at (6,2).

Hence, $x = 6$ and $y = 2$



2. Solve the equation $\frac{2}{x} + \frac{6}{y} = 3$ and $\frac{10}{x} - \frac{9}{y} = 2$ by elimination method.

Solution:

Here,

The given equation are

$$\frac{2}{x} + \frac{6}{y} = 3 \quad \dots\dots\dots (i) \times 3$$

$$\frac{10}{x} - \frac{9}{y} = 2 \quad \dots\dots\dots (ii) \times 2$$

Now, multiplying equation (i) by 3 and equation (ii) by 2 and adding them.

$$\begin{array}{rcl} \frac{6}{x} & + & \frac{18}{y} = 9 \\ \frac{20}{x} & - & \frac{18}{y} = 4 \\ \hline \frac{26}{x} & & = 13 \end{array}$$

$$\text{or, } 13x = 26 \quad \therefore x = 2$$

Again, putting the value of x in equation (i), we get

$$\frac{2}{2} + \frac{6}{y} = 3$$

$$\text{or, } \frac{6}{y} = 3 - 1$$

$$\text{or, } 2y = 6 \quad \therefore y = 3$$

Hence, x = 2 and y = 3

3. Solve the given system of equation by substitution method.

$$\frac{x-1}{y+1} = \frac{1}{2} \text{ and } \frac{x-2}{y+2} = \frac{1}{3}$$

Solution:

Here,

The given equation are

$$\frac{x-1}{y+1} = \frac{1}{2} \quad \text{or, } 2x - 2 = y + 1 \quad \therefore y = 2x - 3 \dots\dots (i)$$

$$\text{and } \frac{x-2}{y+2} = \frac{1}{3} \quad \text{or, } 3x - 6 = y + 2 \quad \therefore 3x - y = 8 \dots\dots (ii)$$

Now, substituting the value of y from equation (i) in equation (ii), we get

$$3x - (2x - 3) = 8 \quad \therefore x = 5$$

Again, Substituting the value of x in equation (i), we get

$$y = 2 \times 5 - 3 = 7$$

Hence, x = 5 and y = 7

4. The cost of 4 kg of chicken and 5 kg of mutton is Rs 7,200. If the cost of 4 kg of chicken is the same as the cost of 1 kg of mutton, find the rate of cost of chicken and mutton.

Solution:

Let the cost of chicken be Rs x per kg and that of the mutton be Rs y per kg.

Then,

According to the given first condition,

$$4x + 5y = 7200 \quad \dots\dots\dots (i)$$

According to the given second condition,

$$4x = y$$

Now, Substituting the value of y in equation (i) from equation (ii). We get

$$4x + 5 \times 4x = 7200$$

$$\text{or, } 24x = 7200 \quad \therefore x = 300$$

Again Substituting the value of x in equation (ii). We get

$$4 \times 300 = y \quad \therefore y = 1200$$

Hence, the rate of cost of chicken is Rs 300 and that of mutton is Rs 1200.

5. **Mother is three times as old as her daughter. Three years ago she was four times as old as her daughter was. Find their present ages.**

Solution:

Let the present age of the mother be x years and that of her daughter be y years

Then,

From the 1st condition;

$$x = 3y \quad \dots (i)$$

From the 2nd condition;

$$x - 3 = 4(y - 3) \quad \dots (ii)$$

Now, Substituting the value of x from equation (i) in equation (ii). We get

$$3y - 3 = 4y - 12 \quad \therefore y = 9$$

Again substituting the value of y in equation (i). We get,

$$x = 3 \times 9 = 27$$

Hence, the present age of the mother is 27 years and that of daughter is 9 years.

6. **The sum of the digits of a two digit number is 10. If 18 is subtracted from the number, the places of the digits are reversed. Find the number.**

Solution:

Let the digit at tens place be x and at ones place be y .

Then, the number is $10x + y$

When the digits are reversed, the new number is $10y + x$

From the first condition,

$$x + y = 10 \quad \text{or, } y = 10 - x \quad \dots (i)$$

From the second condition,

$$10x + y - 18 = 10y + x \quad \text{or, } x - y = 2 \quad \dots (ii)$$

Now, Substituting the value of y from equation (i) in equation (ii), We get

$$x - (10 - x) = 2 \quad \text{or, } 2x = 12 \quad \therefore x = 6$$

Again, Substituting the value of x in equation (i). We get

$$y = 10 - 6 = 4$$

Hence, the required number is $10x + y = 10 \times 6 + 4 = 64$

Extra Questions

1. **Solve each pair of simultaneous equation by graphical method.**

i) $x + 2y = 4$ and $3x - y = 5$ [Ans: 2,1]

ii) $2x - y = 17$ and $2x + 3y + 11 = 0$ [Ans: 5, -7]

2. **Solve each pair of simultaneous equations by elimination method.**

i) $x + 3y = 7$ and $3x - y = 11$ [Ans: 4,1]

ii) $\frac{9}{x} - \frac{4}{y} = 1$, $\frac{15}{x} + \frac{2}{y} = 6$ [Ans: 3,2]

3. **Solve each pair of simultaneous equations by substitution method.**

i) $\frac{x-1}{y+1} = \frac{1}{2}$ and $\frac{x-1}{y+1} = \frac{1}{2}$ [Ans: 10,11]

ii) $x + \frac{x-1}{y+1} = 6$ and $y - \frac{x-1}{y+1} = 2$ [Ans: 4,3]